Build a color-sensor test board

Name: ____________________ Team: ____________________ Date: __________

A board for comparing color readings at a fixed position.

Gather these materials

[ ] Checked base with its downward-facing sensor

[ ] Thin card about 30 cm by 12 cm

[ ] Red, white and blue paper squares, each 8 cm wide

[ ] Removable tape, ruler and pencil

Build: steps 1-4

[ ] 1. Tape the three paper squares side by side on the card: red, white and blue. Leave a small gap between them. Keep the paper flat, without shiny tape over the reading area.

[ ] 2. Keep the robot's wheels raised on firm supports. Slide the board under the sensor. Keep every motor stopped for this lesson.

[ ] 3. Mark the board position that puts the center of red under the sensor. Do the same for white and blue. These marks help you line up each test.

[ ] 4. Measure the gap from the sensor face to the board. Ask the teacher to check it against LEGO's guidance. Record the gap; do not assume one height works for every build.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[ ] 5. Use the program below to show the color name. Test red three times, white three times and blue three times. Move the board away and back to its mark between readings.

[ ] 6. Write down all nine readings, even wrong or unknown ones. Unknown means the sensor cannot name the color.

[ ] 7. Change only one thing, such as room lighting or the sensor gap. Repeat all nine readings. Keep the paper and its position the same.

[ ] 8. Choose the setup that gives the most reliable red and white readings. Record the gap and lighting. Keep the same paper for the stopping lesson.

Make the program

Start with all movement stopped.

Repeat 10 times: read the color, show its name, and wait 0.2 seconds.

Stop all movement and end.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your color readings

Paper / setup	Reading 1	Reading 2	Reading 3
Red / first			
White / first			
Blue / first			
Red / changed			
White / changed			
Blue / changed			

A successful build

Nine readings are recorded for each setup.

Only one thing changes in the comparison.

The team can explain the chosen red-and-white setup.

What did you change, and did it help the robot read colors?